

Extreme Networks Uses Avnet Design Services' ASIC Design Centers to Speed Time-to-Market for Inferno Gigabit Ethernet Chipset

Augments internal engineering team to add ASIC engineering expertise

Extreme Networks is an exciting new networking company that helps customers migrate their existing LANs to meet the Layer 3 switching requirements of today's intranets. Leveraging Ethernet, IP and switching, Extreme Networks delivers long-absent simplicity and consistency throughout the entire LAN infrastructure. That's because all Extreme Layer 3 switches share the same common hardware, software and management architecture. The chassis-based BlackDiamond and stackable Summit switches deliver record-breaking IP routing performance, thanks to ultra-fast custom ASICs and a common non-blocking design.

Extreme Networks was faced with a typical engineering problem: Insufficient internal resources and expertise to get to market quickly. "We were designing our next generation chipset of 5 LSI ASICs and we needed to quickly extend our design team. Our small engineering group focused all their attention on the architectural definition, system level issues and intellectual property (IP) creation tasks. To keep our productivity at a maximum, and accomplish all the implementation tasks -- things like floor-planning, block integration, test vector generation, scan insertion, and verification -- we needed to grow the team by 4 times our size," explained Curt Berg, ASIC Design Manager for Extreme Networks. "We also needed to add these capabilities without adding additional computing resources, training or a long interview and hiring process. That's where Avnet Design Services came in."

"Avnet Design Services not only had experts who could solve all of our integration, testing and verification questions, but they had access to the key LSI Logic engineers who could provide quick answers to detailed chip level questions, about things like process limitation and SRAM cell design. In a few cases this was critical to keeping our project on track," continued Curt.

"Our resource profile is not unusual in this industry. There are only spikes in our resource requirements, no smooth ramps up or down or steady states. Avnet Design Services experts helped us out at just the right

time with just the right amount of resources -- knowledgeable engineers with the tools and experience we required."

"We worked together as a team, one team per ASIC, holding weekly meetings to review progress and address issues. We were able to use email and remote file transfer, allowing team members to work easily from remote locations. On several occasions, multiple design centers were working on our project simultaneously. The additional resources kept us on schedule, and Avnet Design Services was able to bring in designers from Chicago or Boston as easily as from San Jose. ADS has the expertise, the tools and the support from LSI Logic we needed to get our ASICs completed in record time."

The high-density Summit7i™ LAN switch delivers non-blocking wire-speed IP/IPX routing and switching to server farms, aggregated switches and network backbones. Available with 28 100/1000BASE-T or 1000BASE-SX ports, plus four GBIC-based 1000BASE-X ports, Summit7i is the industry's first stackable 32-port Gigabit Ethernet switch in a compact E4U• package.

The high port-density of the Summit7i makes it easy and cost-effective to scale backbone bandwidth and aggregate multiple switches. As a mid-tier switching solution, Summit7i can aggregate multiple Summit24™ and Summit48™ Enterprise Desktop Switches, while providing high-speed gigabit links to BlackDiamond™ chassis switches in the core. The core architecture of the Summit products can be traced to the 'Inferno' ASIC chipset—a common building block and the heart of the family's wire speed performance feature.

"The Inferno chipset has 5 ASICs total. There are between .5M and 1M gates and 10-20 SRAM blocks per chip. This chipset is the backbone for our next generation of Ethernet Switch product lines. It is used in the Summit7i product and on the blades for the Black Diamond chassis," explained Curt.

"ADS has the expertise, the tools and the support from LSI Logic we needed."

*Curt Berg,
ASIC Design Manager
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